

GDP Growth and Economic Performance: Kenya in East African and Global Context (2015–2025)

AnalystLab Africa — Week 8 Data Analytics Capstone Project

Prepared by: Tess Kamau

Data Source: World Bank — World Development Indicators (WDI)

Tools: Python (pandas, matplotlib) · Power BI

This report documents the complete, end-to-end analytics workflow behind the accompanying interactive Power BI dashboard — from raw data acquisition through cleaning, exploratory analysis, visualization, and final recommendations.

1. Objective / Problem Statement

Kenya's economic trajectory is best understood not in isolation, but in comparison with its regional peers and the wider global economy. This project analyzes GDP growth and related macroeconomic indicators for Kenya alongside six comparator African economies — spanning small East African neighbours to the continent's two largest economies — plus Sub-Saharan Africa and World benchmarks, over an eleven-year window (2015–2025).

The objective is to identify how Kenya's growth, price stability, and wealth-per-person compare to this peer set; to surface any notable shocks, turning points, or structural patterns in the data; and to translate these findings into an interactive Power BI dashboard and a set of clear, evidence-based insights and recommendations for stakeholders such as policymakers, investors, and development analysts.

2. Dataset Description

The dataset is sourced from the World Bank's **World Development Indicators (WDI)**, the World Bank's flagship collection of global development statistics, covering over 200 countries and territories across economic, social, and environmental themes. The full bulk-download extract (*WDICSV.csv*, ~189MB) contains 396,970 rows and 70 columns, representing every country/indicator combination tracked by the World Bank, with one column per year from 1960 to 2025.

For this project, the dataset was narrowed to:

- **9 geographic units:** Kenya (focus country), Burundi, Nigeria, Rwanda, South Africa, Tanzania, Uganda, plus two benchmarks — Sub-Saharan Africa and World.
- **5 indicators:** GDP (current US\$), GDP growth (annual %), GDP per capita (current US\$), Inflation – consumer prices (annual %), and Exports of goods and services (% of GDP).
- **11 years:** 2015 through 2025, chosen because this window is fully populated for nearly every indicator/country combination, avoiding the sparse coverage found in earlier decades.

3. Data Cleaning Process

All cleaning and transformation was performed in Python (pandas) inside a documented Jupyter notebook. The process followed these steps:

- **Filtering:** The raw 396,970-row dataset was filtered down to the 9 target countries and 5 target indicators, producing 45 rows (9×5).
- **Reshaping (melt):** The data was originally in wide format, with a separate column for every year since 1960. It was trimmed to the 2015–2025 range and melted into long/tidy format (one row per country–indicator–year), producing 495 rows.
- **Pivoting:** The long-format data was then pivoted so that each of the 5 indicators became its own column, with one row per country per year — the final analysis-ready structure (99 rows $\times$ 8 columns: 9 countries $\times$ 11 years).

- **Missing value check:** A completeness audit found 22 missing values, all confined to the "Exports of goods and services (% of GDP)" indicator for **Burundi** and **Nigeria** across every year in the range — a genuine gap in the World Bank's source data for these two countries, not an artifact of the cleaning process.
- **Handling decision:** These 22 values were left as missing (NaN) rather than estimated or filled with zero, since either alternative would misrepresent the underlying reality. This is documented transparently rather than concealed, and is visible in the dashboard as a data gap for those two countries on that specific indicator.
- **Currency-year caveat:** 2025 figures are the most recent available in this WDI extract but may still be subject to future revision, consistent with the World Bank's typical 1–2 year finalization lag.

4. Analysis Methodology

Analysis proceeded in two phases. First, exploratory analysis was carried out in Python to identify the dataset's key patterns before any dashboard was built — ensuring the final Power BI report was designed around genuine findings rather than built speculatively. Second, these same indicators were rebuilt as an interactive Power BI dashboard, allowing any viewer to explore the same patterns by country and year.

Python exploratory techniques used:

- Grouped summary statistics (mean GDP growth, inflation, and GDP per capita by country, 2015–2025).
- Single-country time series analysis (Kenya's year-by-year trajectory).
- Multi-country comparative line charts (GDP growth and inflation over time, all countries overlaid).
- Cross-sectional ranking (GDP per capita, most recent year).
- Paired-metric comparison (average growth vs. average inflation per country).

The Power BI dashboard mirrors these same analytical views but adds interactivity: Country and Year slicers drive four headline KPI cards (GDP Growth, GDP per Capita, Inflation, Total GDP), while three always-on comparison visuals (two trend lines and one ranking bar chart) give a constant, unfiltered view of how all nine units compare across the full time period.

5. Key Findings

5.1 Country Summary, 2015–2025 Averages

Country	Avg GDP Growth (%)	Avg Inflation (%)	Avg GDP/Capita (US\$)
Rwanda	7.02	6.88	881
Tanzania	5.31	4.05	1,108
Uganda	4.97	4.27	907
Kenya	4.63	6.02	1,913
Burundi	3.08	12.68	248
World	2.82	3.20	11,979
Sub-Saharan Africa	2.44	4.77	1,659
Nigeria	1.49	17.70	2,246
South Africa	0.74	4.86	6,334

Kenya's row is highlighted. Source: author's calculations from WDI, 2015–2025.

- **Rwanda leads on growth** (7.02% avg) but also carries the second-highest inflation (6.88%) and the lowest GDP per capita — fast growth from a small economic base.

- **Kenya sits mid-pack on growth** (4.63%, above both the World and Sub-Saharan Africa averages) but holds the **highest GDP per capita among its immediate East African peers** (\$1,913) — well ahead of Rwanda, Tanzania, Uganda, and Burundi.
- **Nigeria** combines very low growth (1.49%) with by far the highest inflation (17.70%) — a stagflation-adjacent profile.
- **South Africa** shows the lowest growth of all (0.74%) despite the highest GDP per capita (\$6,334) — consistent with a mature, stagnating economy.
- **Burundi** is the clear low-end outlier on wealth, with GDP per capita of just \$248, a fraction of every other country in the set.

5.2 Kenya's Trajectory, 2015–2025

Kenya's year-by-year path tells a coherent recovery story. Growth was steady through 2015–2019 (3.8–5.6%), collapsed to **-0.27%** in 2020 — the only negative year in the series, driven by the COVID-19 shock — then rebounded sharply to **7.59%** in 2021, the strongest single year in the whole dataset. Inflation spiked to its highest points (7.66% and 7.67%) in 2022–2023, likely reflecting global commodity and currency pressures, before cooling to 4.49% and 4.07% in 2024–2025 — the healthiest growth-with-controlled-inflation combination in the entire series, alongside Kenya's highest recorded GDP per capita figures (~\$2,133 and ~\$2,363).

5.3 GDP Growth: Kenya vs. Peer Countries

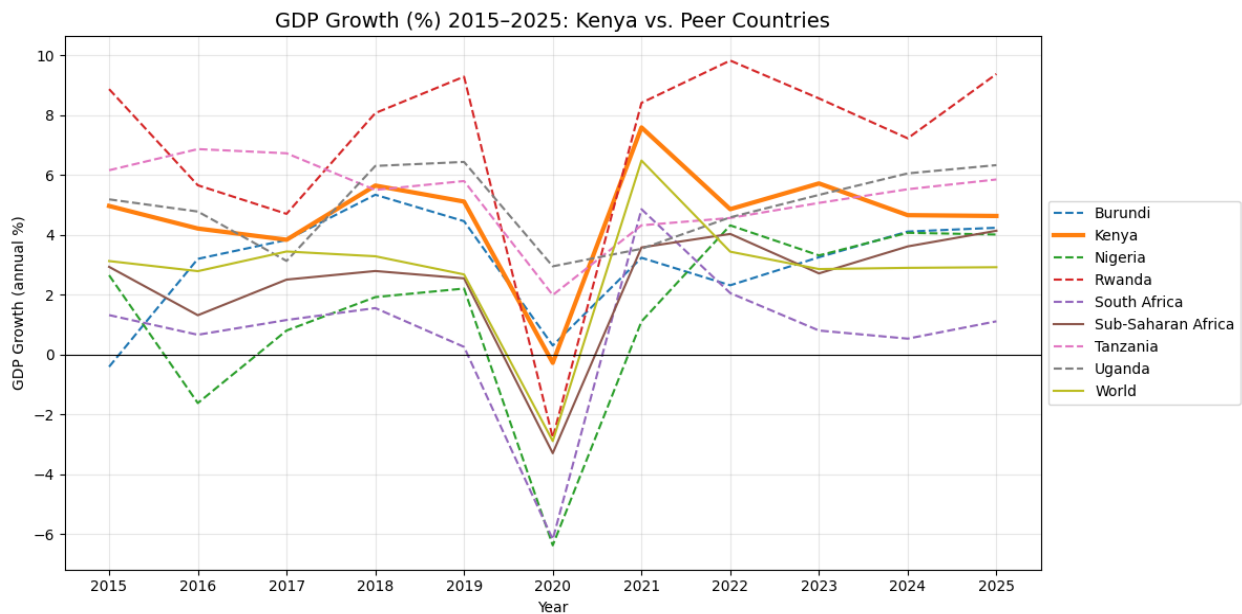


Figure 1. GDP growth (%), 2015–2025, Kenya vs. peer countries.

The 2020 shock affected nearly every country, but unevenly: Nigeria, South Africa, and Sub-Saharan Africa as a whole fell to roughly -6% to -3% growth, while Kenya's dip was far shallower (~-0.3%) — suggesting **Kenya's economy was notably more resilient to the COVID-19 shock** than most peers, plausibly reflecting a less commodity/export-dependent structure than oil-reliant Nigeria or South Africa. The 2021 rebound was sharp across the board, but Rwanda and Kenya posted among the strongest recoveries. Rwanda's line is consistently the most volatile and highest-peaking, while South Africa's is the flattest throughout — visual confirmation of a stagnating, mature economy.

5.4 GDP per Capita Ranking, 2025

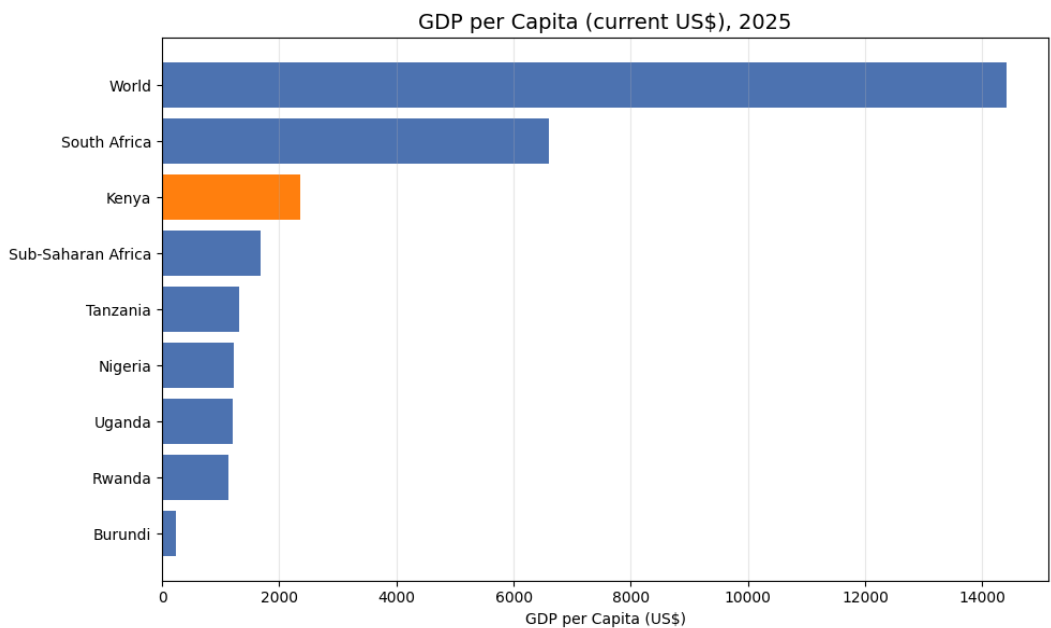


Figure 2. GDP per capita (current US\$), 2025 ranking.

Kenya ranks third overall in 2025 — behind only the World average and South Africa, and clearly ahead of Sub-Saharan Africa, Tanzania, Nigeria, Uganda, Rwanda, and Burundi. The most striking finding here is **Nigeria**: despite having the second-highest average GDP per capita across 2015–2025, it had fallen to near the bottom of the group by 2025. Quantifying this directly, Nigeria's GDP per capita **fell 52.7% between 2015 and 2025** — from \$2,585.73 to \$1,224.25. The decline was not gradual: it fluctuated through 2015–2023, then collapsed sharply in 2024 to \$1,084.16 (nearly halving in a single year) before a slight recovery in 2025. The timing aligns closely with Nigeria's naira devaluation and FX reforms in mid-2023, which sharply reduced dollar-denominated GDP per capita independent of underlying local growth — illustrating that GDP per capita in current US\$ is sensitive to exchange-rate movements, not growth alone.

5.5 Growth vs. Inflation by Country

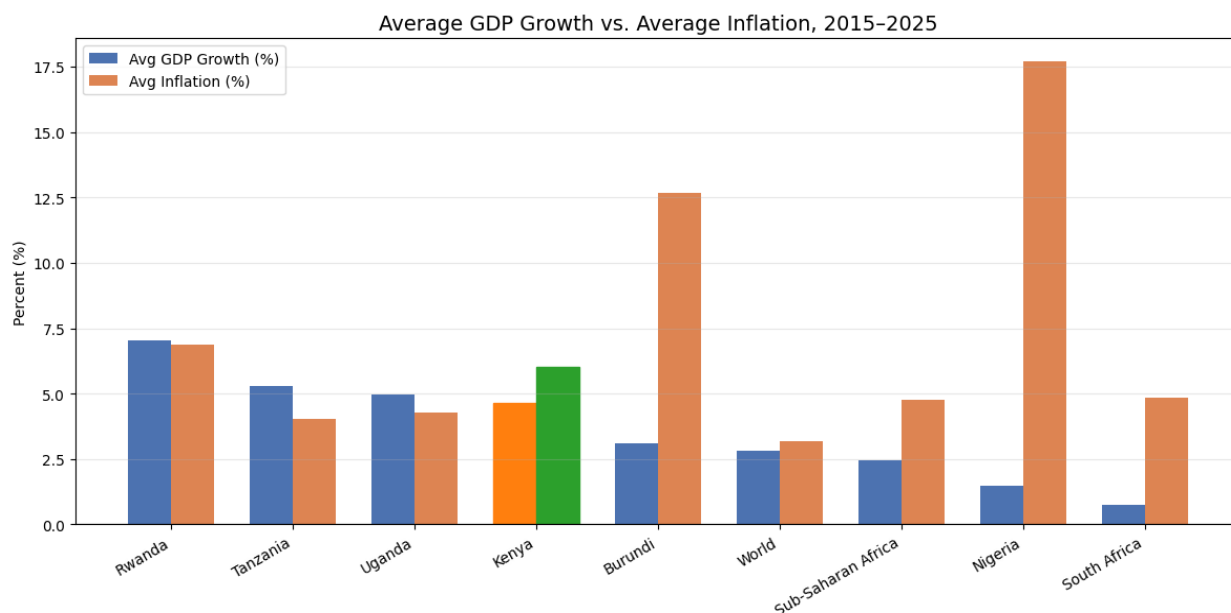


Figure 3. Average GDP growth vs. average inflation by country, 2015–2025.

Comparing average growth to average inflation per country reveals a clear pattern: high-growth countries generally show inflation roughly in line with or below their growth rate (Rwanda, Tanzania, Uganda). **Kenya sits in a healthy position** — 4.63% growth against 6.02% inflation, better balanced than the low-growth, high-inflation countries. Low-growth countries tend to show disproportionately high inflation, most visible in Nigeria (1.49% growth vs. 17.70% inflation) and Burundi (3.08% vs. 12.68%). South Africa is a partial exception — low growth but only moderate inflation — suggesting its stagnation is structural rather than inflation-driven.

6. Insights

- **Kenya is a stable, above-average performer rather than a boom-bust economy.** It does not post Rwanda-level growth spikes, but it also avoided Rwanda-level volatility and Nigeria/South Africa-level shocks, while holding the highest GDP per capita among its immediate East African peers.
- **Kenya's COVID-19 resilience is a genuine competitive strength.** Its 2020 contraction (-0.27%) was far shallower than regional and continental peers, and its 2021 rebound (7.59%) was among the strongest in the dataset.
- **Kenya's 2024–2025 macro position is the healthiest in its own recent history** — steady growth combined with cooling inflation and record GDP per capita — a favourable backdrop for continued investment and policy planning.
- **Nigeria's per-capita wealth collapse is a currency story, not purely a growth story.** Its GDP growth stayed modestly positive through most of the period even as dollar-denominated GDP per capita fell by over half — a reminder that current-US\$ metrics can mask or exaggerate underlying economic reality when currencies move sharply.
- **High inflation, more than low growth alone, appears to separate the weaker performers** (Nigeria, Burundi) from the stronger ones in this dataset — price stability looks like a meaningful

differentiator across this peer group.

- **Rwanda's growth model carries a volatility trade-off** worth further study: it is the fastest grower but also swings the widest, and starts from the smallest per-capita base.

7. Recommendations

For policymakers and analysts focused on Kenya:

- Continue prioritizing inflation management alongside growth — Kenya's best-performing years (2024–2025) combined moderate growth with the lowest inflation in the series, suggesting this balance, not growth alone, is what should be protected and replicated.
- Use Kenya's demonstrated COVID-19 resilience as a evidence-based talking point in investment and development-partner discussions — the shallow 2020 contraction relative to peers is a genuine, data-backed differentiator.
- Monitor exchange-rate exposure proactively. Nigeria's experience shows how quickly a currency shock can erase years of per-capita gains in dollar terms; stress-testing Kenya's own currency exposure would be a reasonable precaution.

For investors and development partners evaluating the region:

- Weigh Rwanda's high growth against its volatility — it may suit longer investment horizons better than short-term positioning.
- Treat Nigeria's per-capita figures with caution until FX stability is clearer, since the 2023–2024 collapse was currency-driven rather than a reflection of a shrinking real economy.
- Consider Kenya's combination of regional wealth leadership, growth stability, and shock resilience as a comparatively lower-volatility entry point into East African markets relative to its peers.

For further analysis:

- Extend the indicator set to include exchange rate and current account data directly, to more precisely separate currency effects from real economic growth (particularly relevant for the Nigeria finding).
- Investigate the Burundi and Nigeria export-data gap with an alternative source, since Exports of goods and services (% of GDP) was unavailable for both across the full period in this dataset.

8. Conclusion

Across 2015–2025, Kenya emerges from this analysis as a comparatively stable, resilient, and steadily improving economy within its East African and continental peer group. It is not the fastest grower — that distinction belongs to Rwanda — but it combines above-average growth, regional wealth leadership, and demonstrated shock resilience with a notably healthier growth-inflation balance than several larger peers, including Nigeria and South Africa. The most striking single finding in the dataset, Nigeria's 52.7% GDP-per-capita decline, underscores an important analytical lesson carried throughout this project: headline dollar-denominated figures can move sharply due to currency effects alone, and any comparative economic analysis should account for that before drawing conclusions about real underlying growth.

This report, together with the accompanying interactive Power BI dashboard, GitHub repository, and demonstration video, represents a complete, end-to-end data analytics workflow — from raw World Bank data through cleaning, exploratory analysis, visualization, and actionable recommendations — completed as the final capstone project of the AnalystLab Africa Data Analyst Internship.

Appendix: Dashboard Overview

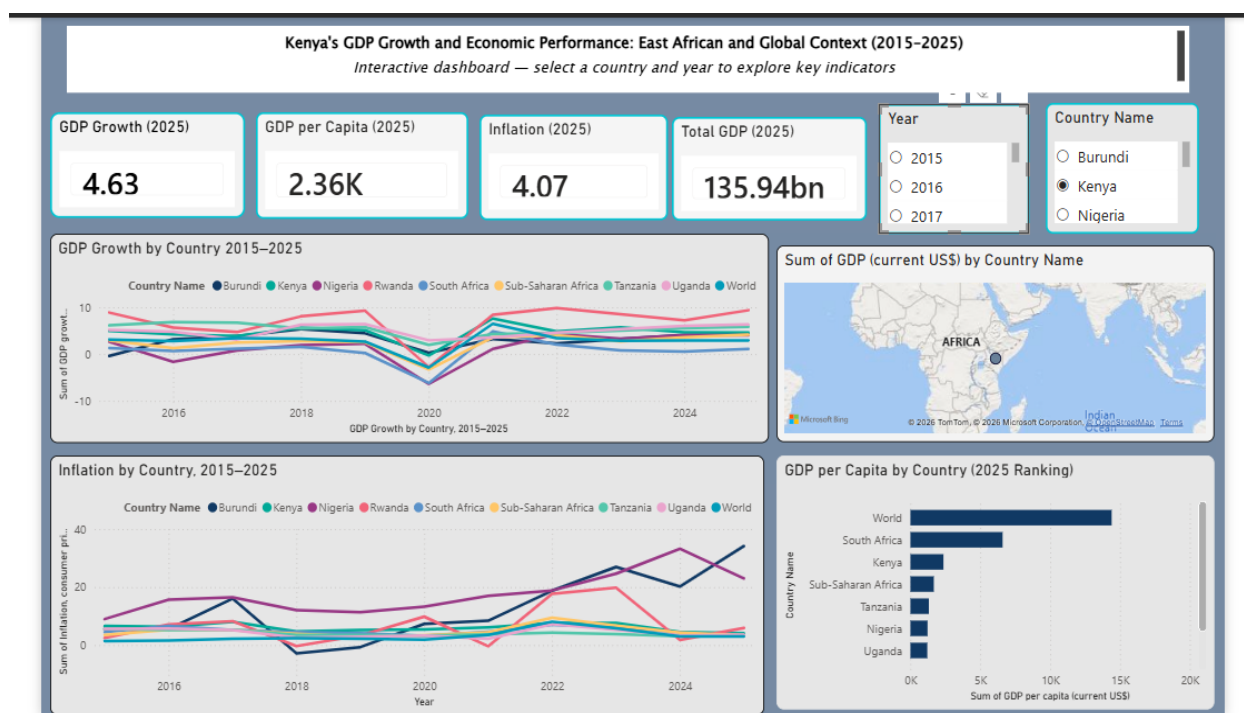


Figure 4. Final interactive Power BI dashboard — Kenya's GDP Growth and Economic Performance in East African and Global Context (2015–2025).